Title:

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$f(t)$ 与奇异信号卷积

① $f(t) * \delta(t) = f(t)$

② $f(t) * \delta(t-t_0) = f(t-t_0)$

③ $f(t) * \delta'(t) = f'(t)$

④ $f(t) * \delta^{(n)}(t) = f^{(n)}(t)$

⑤ $f(t) * u(t) = \int_{-\infty}^{t} f(\tau)d\tau$

⑥ $f(t) * t u(t) = \iint f(t)$

① $f[n] * \delta[n] = f[n]$

② $f[n] * \delta[n-n_0] = f[n-n_0]$

③ $f[n] * u[n] = \sum_{m=0}^{+\infty} f[n-m] = \sum_{m=-\infty}^{n} f(m)$

斜坡信号：$t u(t) = \int_{-\infty}^{t} u(\tau)d\tau$

$u(t) * u(t) = t u(t)$

$u(t) * u(-t)$ 不存在

卷积的时移特性

$f_1(t) * f_2(t) = f(t)$

$f_1(t) * f_2(t-t_0) = f(t-t_0)$

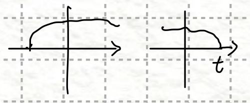
$f_1(t-t_0) * f_2(t-t_1) = f(t-t_0-t_1)$

$f_1[n] * f_2[n] = f[n]$

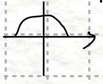
$f_1[n-n_0] * f_2[n-n_1] = f[n-n_0-n_1]$

卷积性质可应用的条件、

①信号有始，有终



②其中一个时限的



直流

$(A \neq 0)$

$A * u(t)$ 不存在

$u(-t) * u(t)$

$f(t)$ 当 $t=-\infty$ 时 不为0

$f(t) * u(t)$

卷积的微积分特性

$f'(t) = f_1'(t) * f_2(t)$

$= f_1(t) * f_2'(t)$

若 $f_1(t) * f_2(t) = f(t)$

则 $f_1(t) * \frac{df_2(t)}{dt} = \frac{df(t)}{dt}$

交换律合 一样适用

$f_1(t) * \int_{-\infty}^{t} f_2(\tau)d\tau = \int_{-\infty}^{t} f(\tau)d\tau$

$\frac{df_1(t)}{dt} * \int_{-\infty}^{t} f_2(\tau)d\tau = f(t)$

$f_1^{(k)}(t) * f_2^{(-r)}(t) = f^{(k-r)}(t)$

$f_1^{(r)}(t) * f_2^{(k)}(t) = f^{(k+r)}(t)$

$f_1^{(-r)}(t) * f_2^{(-k)}(t) = f^{(-(k+r))}(t)$

（$f_2^{(-k)}(t)$ 表示对 $f_2(t)$ 进行 k 次变上限积分

k、r 均为正整数）

$f^{(i)}(t) = f_1^{(j)}(t) + f_2^{(i-j)}(t)$

i、j 可 +、−

↓微分 ↓积分

卷积的尺度变换

$f_1(t) * f_2(t) = f(t)$

$f_1(at) * f_2(at) = \frac{1}{|a|} f(at) \quad (a \neq 0)$

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常见的卷积

卷积求法：

1) 定义

2) 图解

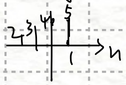
3) 性质

①有限长序列的卷积

对位相乘求和法，

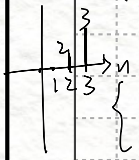
(1)

(2) a+c　b+d
　　左　　右



$*$

$\{2\ 3\ 4\ 5\}$



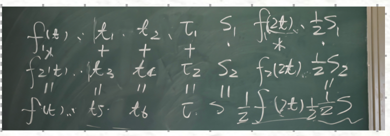
$\{0\ 0\ 2\ 3\}$

```
    2 3 ④ 5
    ⓪ 0 2 3        不进位
  ----------------
    6 9 12 15
  4 6 8 10
0 0 0 0
0 0 ⓪ 0
----------------
  4 12 17 22 15
```

$y(n)=\{4\ 10\ 12\ 22\ 15\}$

n=0.

②两矩形信号卷积



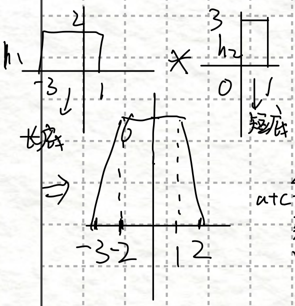
$f_1(t)$、t_1、t_2、τ_1　S_1　$f_1(2t)$、$\frac{1}{2}S_1$

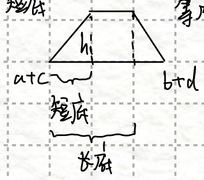
$*$　+　+　+　·　$*$　·

$f_2(t)$、t_3　t_4　τ_2　S_2　$f_2(2t)$　$\frac{1}{2}S_2$

"　"　"　"　"　"

$f(t)$　t_5　t_6　τ　S　$\frac{1}{2}f(2t)$　$\frac{1}{2}\cdot\frac{1}{2}S$

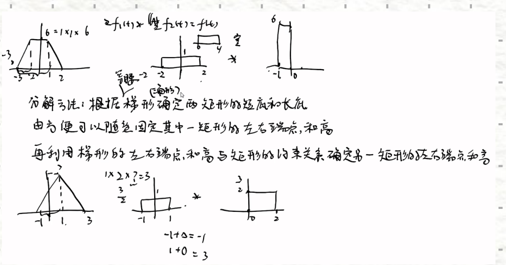




等腰

$h=h_1\cdot h_2\cdot$短底

也可由△或⏢分解为两个矩形



分解方法：根据梯形确定两矩形的短底和长底

由高便可以随意固定其中一矩形的左右端点，和高

再利用梯形的左右端点，和高与矩形的的来关系确定另一矩形的左右端点和高

③常数卷积

$f(t)*A=$ A倍的 $f(t)$ 的净面积（微积分求 $\int$ 上下拼凑）

④一些卷积积分/积

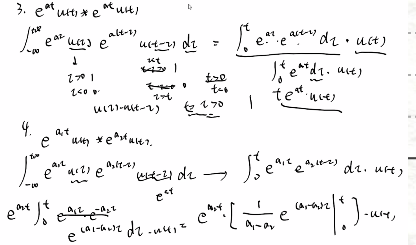
本科课程用得少
暂缓

1. $U(t)*U(t)=t\,U(t)$

$\int_{-\infty}^{t}f(\tau)U(\tau)d\tau=\int_{0}^{t}f(\tau)d\tau\cdot u(t)$

（$U(\tau)$提到$\int$外面）

2. $e^{at}\cdot U(t)*U(t)=\frac{1}{a}(e^{at}-1)\cdot U(t)$



3. $e^{at}u(t)*e^{at}u(t)$

$\int_{-\infty}^{+\infty}e^{a\tau}\underline{u(\tau)}\,e^{a(t-\tau)}u(t-\tau)\,d\tau=\underline{\int_{0}^{t}e^{a\tau}\cdot e^{a(t-\tau)}d\tau}\cdot\underline{u(t)}$

$\tau>0$ 1
$\tau<0$ 0

$\tau<t$ ~~$t-\tau>0$~~ 1
~~$t-\tau<0$~~ 0
$\tau>t$

$u(\tau)-u(t-\tau)$

$\frac{t>0}{t<0}$

$t\geq\tau>0$　1

$\int_{0}^{t}e^{at}d\tau\cdot u(t)$

$\underline{te^{at}\cdot u(t)}$

4. $e^{a_1t}u(t)*e^{a_2t}u(t)$

$\int_{-\infty}^{+\infty}e^{a_1\tau}u(\tau)\,e^{a_2(t-\tau)}u(t-\tau)\,d\tau\longrightarrow\int_{0}^{t}e^{a_1\tau}e^{a_2(t-\tau)}d\tau\cdot u(t)$

e^{ct}

$e^{a_2t}\int_{0}^{t}\underline{e^{a_1\tau}\cdot e^{-a_2\tau}}\,d\tau\cdot u(t)=e^{a_2t}\cdot\left[\frac{1}{a_1-a_2}e^{(a_1-a_2)\tau}\Big|_{0}^{t}\right]\cdot u(t)$

$e^{(a_1-a_2)\tau}$

此处均为
定义推得

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